

Predicting Intra-Treatment Complications During Continuous Renal Replacement Therapy Using MIMIC-IV

Study Protocol — Version 0.1

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1. Background and Gap

Prediction work on public ICU data has concentrated on three CRRT endpoints — mortality after initiation, prediction of which patients will require CRRT, and weaning success — all of which are well covered. **What has not been modeled at scale is what goes wrong during therapy.** MIMIC-IV records hourly circuit telemetry (access, filter, effluent and return pressures; blood flow, dialysate, replacement and ultrafiltration rates) that most public databases lack, while prior work on circuit clotting is limited to single-center cohorts of roughly 100–650 patients.

2. Objectives

- **Primary** Develop and internally validate a model predicting premature CRRT circuit failure due to clotting within a 6-hour horizon.
- **Secondary** Predict incident severe hypophosphatemia during CRRT within 24 hours.
- **Tertiary** Quantify the incremental value of machine-level circuit features over routine clinical and laboratory variables (group ablation).

3. Data Source and Governance

MIMIC-IV (PhysioNet, credentialed access). CITI “Data or Specimens Only Research” completed; credentialing under faculty reference; DUA signed per version; each investigator credentialed individually with no data sharing between them. Pipeline developed against the open-access MIMIC-IV Demo. Institutional non-human-subjects determination to be obtained. Code public; data never redistributed.

4. Study Design and Cohort

Retrospective cohort, single center. **Include** adults ≥ 18 years at ICU admission with ≥ 1 documented CRRT session of ≥ 4 hours. **Exclude** chronic dialysis dependence prior to admission; CRRT documented without machine parameters; comfort-measures-only before CRRT onset. **Unit of analysis** the CRRT circuit; patients contribute multiple circuits. Sessions are constructed by stitching consecutive hourly machine rows, with a >2 -hour gap defining a new session (sensitivity: 1h, 4h). Train/test splits are made *by patient*, never by circuit. STROBE flow diagram maintained with counts at each exclusion step.

5. Outcome Definition

Circuit terminations are extracted with their documented reason and classified as clotting-related, elective/scheduled, patient-related (death, transport, procedure), or unclear, corroborated by terminal pressure trajectory and by circuit lifespan relative to the scheduled change interval. **A random sample of ~ 150 terminations will be manually adjudicated by a clinical co-investigator**, with agreement reported as Cohen’s κ ; primary analysis proceeds only if $\kappa \geq 0.6$. Unclear cases are excluded in the primary analysis and treated as non-events in sensitivity analysis. Hypophosphatemia is defined as serum phosphate < 2.0 mg/dL (severe: < 1.0), incident among circuits above threshold at prediction time, with phosphate repletion modeled as a competing intervention.

6. Prediction Task and Leakage Control

At each hour t of an active circuit, using only data available at or before t , predict whether the event occurs in $(t, t + H]$, with $H = 6$ h primary and 3h/12h as sensitivity. A 30–60 minute blanking interval immediately preceding the event is excluded from the feature window. Circuits are scored only after ≥ 2 hours of runtime. Imputation and scaling parameters are fit inside training folds only. A shuffled-label negative control is run on every pipeline change.

7. Candidate Predictors

Circuit telemetry with engineered trends (1h/3h/6h last value, mean, min, max, slope, variance; blood-flow-adjusted pressure drop; ultrafiltration-adjusted transmembrane pressure); anticoagulation strategy and dose, including citrate and total/ionized calcium ratio; coagulation and hematology; hemodynamics and vasopressor dose; chemistry; vascular access site; demographics, comorbidity and severity scores at CRRT onset; measurement-presence and time-since-last-measurement indicators.

8. Analysis, Validation and Evaluation

Model ladder: (i) published two-parameter pressure rule as clinical baseline, (ii) penalized logistic regression, (iii) gradient boosting, (iv) sequence model only if event volume permits. Nested cross-validation grouped by patient. Calibration by Platt or isotonic scaling fit within training folds. **Validation** grouped k -fold with bootstrap optimism correction, plus temporal validation across MIMIC-IV admission-year groups; external validation attempted on a reduced feature set. **Metrics** AUPRC with stated prevalence (primary), AUROC, calibration slope and intercept, Brier score; event-level rather than row-level sensitivity; PPV and number-needed-to-alert at a fixed alert rate; median lead time; decision curve analysis; subgroup performance by CRRT mode, anticoagulation strategy, sex, and age.

9. Open Methodological Questions

- Circuits are clustered within patients: preferred approach to variance estimation and to preventing optimistic performance estimates from within-patient correlation?
- Circuit failure is time-to-event with competing risks (death, elective discontinuation). Do circuits still running at end of follow-up constitute length bias, or is left truncation the correct framing?
- Is a fixed-horizon binary formulation defensible, or should the primary analysis be a competing-risks survival model with a landmark structure?

10. Reporting and Reproducibility

TRIPOD+AI checklist completed at design stage; PROBAST+AI self-assessment before submission; protocol pre-registered on OSF prior to outcome data inspection. Public repository with extraction SQL, feature pipeline, models, pinned environment, data dictionary and decision log.